Phase 1: Research Summary – Currency API, Visualization & Security

1. Currency Conversion API

Selected: Open Exchange Rates

- Provides a developer-friendly JSON API for live and historical exchange rates for over 200 currencies, trusted by 100,000+ developers
- Clean JSON responses with transparent documentation
- Open-source integrations (200+ community-built libraries/plugins)
- Free tier available

2. Visualization Library

Chosen: Recharts

- Built for React: Seamless integration with JSX and React components
- Declarative & composable: complex charts by composing simple components
- Extensive chart types: Bar, line, pie, area, scatter, and more — ideal for expense and budget visuals

3. Security & Best Practices

- **Environment & Secret Management**: Keep API keys in backend .env files.
- **HTTPS/TLS**: Enforce TLS (HTTPS) for all client-server and API interactions to maintain secure transport.
- **Input Validation**.
- **Authentication**: Use per-user auth, and consider multi-factor auth.
- **Secure Data Storage**: Encrypt sensitive data at rest

✅ Summary & Next Steps

- **Decisions made**: Open Exchange Rates + Recharts for currency conversion and analytics; robust app security foundations established.
- **Upcoming**: Prepare environment, register API key, and begin Phase 2 (architecture design, wireframes, DB schema).